Practical 10

WINE DATASET ANALYSIS AND PRINCIPAL COMPONENT ANALYSIS (PCA)

Problem statement-

1.Display the Basic Statistics using describe ()

2. Boxplot by output labels/classes

3. Scatterplot using two variable

4. Plot Co-Variance Matrix

5. Data Scaling

6. Show better class separation using principal components

Solution - here is the complete code

```
# Import necessary libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

import warnings
warnings.filterwarnings('ignore')

print("All libraries imported successfully!")
```

```
All libraries imported successfully!
```

```
# STEP 1 - BASIC STATISTICS

wine = load_wine()

# Convert dataset into DataFrame
data = pd.DataFrame(wine.data,columns=wine.feature_names)
# Add target class
data["Class"] = wine.target

# Convert class numbers into class names
class_names = {index: name for index, name in enumerate(wine.target_names)}
data["Class_Name"] = data["Class"].map(class_names)

# Display dataset information
print("Dataset loaded successfully!")
print("Dataset shape:", data.shape)

print("\nFirst five rows:")
display(data.head())

print("\nBasic statistical description:")
display(data[wine.feature_names].describe().round(2))

print("\nNumber of samples in each class:")
print(data["Class_Name"].value_counts().sort_index())
```

```
Dataset loaded successfully!
Dataset shape: (178, 15)

First five rows:
```

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	proanthocyanins	c
0	14.23	1.71	2.43	15.6	127.0	2.80	3.06	0.28	2.29	
1	13.20	1.78	2.14	11.2	100.0	2.65	2.76	0.26	1.28	
2	13.16	2.36	2.67	18.6	101.0	2.80	3.24	0.30	2.81	
3	14.37	1.95	2.50	16.8	113.0	3.85	3.49	0.24	2.18	
4	13.24	2.59	2.87	21.0	118.0	2.80	2.69	0.39	1.82	

```
Basic statistical description:
```

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	proanthocyan
count	178.00	178.00	178.00	178.00	178.00	178.00	178.00	178.00	178
mean	13.00	2.34	2.37	19.49	99.74	2.30	2.03	0.36	1
std	0.81	1.12	0.27	3.34	14.28	0.63	1.00	0.12	0
min	11.03	0.74	1.36	10.60	70.00	0.98	0.34	0.13	0
25%	12.36	1.60	2.21	17.20	88.00	1.74	1.20	0.27	1
50%	13.05	1.87	2.36	19.50	98.00	2.36	2.13	0.34	1
75%	13.68	3.08	2.56	21.50	107.00	2.80	2.88	0.44	1
max	14.83	5.80	3.23	30.00	162.00	3.88	5.08	0.66	3

```
Number of samples in each class:
Class_Name
class_0    59
class_1    71
class_2    48
Name: count, dtype: int64
```

Interpretation:

The Wine dataset contains 178 observations and 13 numerical features. The describe() function provides statistical information such as count, mean, standard deviation, minimum, maximum and quartiles for each feature. The dataset is divided into three classes: Class 0 contains 59 samples, Class 1 contains 71 samples, and Class 2 contains 48 samples. This analysis helps understand the structure, distribution and statistical properties of the dataset.

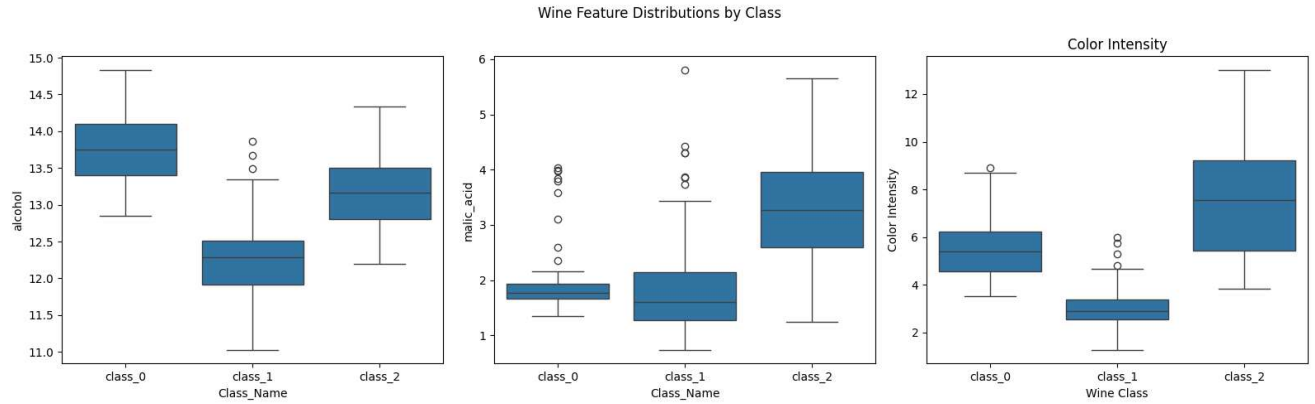
```
# STEP 2 - BOXPLOTS BY CLASS

selected_features = ["alcohol","malic_acid","color_intensity"]
fig, axes = plt.subplots(1, 3, figsize=(16, 5))
for ax, feature in zip(axes, selected_features):
 sns.boxplot(
        data=data,
        x="Class_Name",
        y=feature,
        ax=ax
    )

ax.set_title(feature.replace("_", " ").title())
ax.set_xlabel("Wine Class")
ax.set_ylabel(feature.replace("_", " ").title())

plt.suptitle("Wine Feature Distributions by Class")
plt.tight_layout()
plt.show()

print("Boxplots generated successfully!")
```




```
Boxplots generated successfully!
```

Interpretation:

The boxplots represent the distribution of alcohol, malic acid and color intensity across the three wine classes. They show the median, quartiles, spread and potential outliers of each feature. The differences in boxplot distributions indicate that these features have varying characteristics across different wine classes. This visualization helps compare the variation of selected features among the three classes.

```
# STEP 3 - SCATTERPLOT

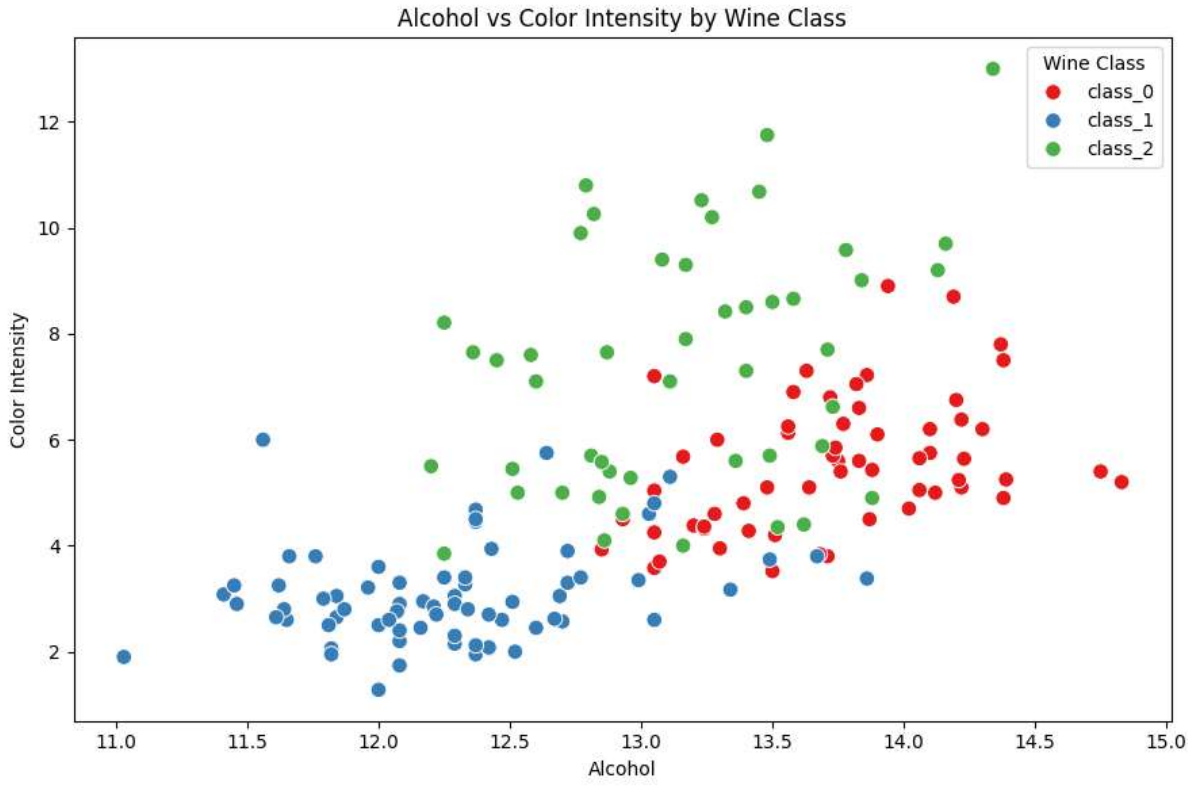
plt.figure(figsize=(9, 6))

sns.scatterplot(
    data=data,
    x="alcohol",
    y="color_intensity",
    hue="Class_Name",
    palette="Set1",
    s=70
)

plt.title("Alcohol vs Color Intensity by Wine Class")
plt.xlabel("Alcohol")
plt.ylabel("Color Intensity")
plt.legend(title="Wine Class")

plt.tight_layout()
plt.show()

print("Scatterplot generated successfully!")
```




```
Scatterplot generated successfully!
```

Interpretation:

The scatterplot illustrates the relationship between alcohol and color intensity for the three wine classes. Alcohol is represented on the X-axis, while color intensity is represented on the Y-axis. Different colours indicate different wine classes. The plot helps identify patterns, relationships and differences between the classes based on these two numerical features.

```
# STEP 4 - COVARIANCE MATRIX

features = data[wine.feature_names]

# Calculate covariance matrix
covariance_matrix = features.cov()

print("Covariance Matrix:")
display(covariance_matrix.round(2))

# Improved covariance heatmap
plt.figure(figsize=(14, 10))

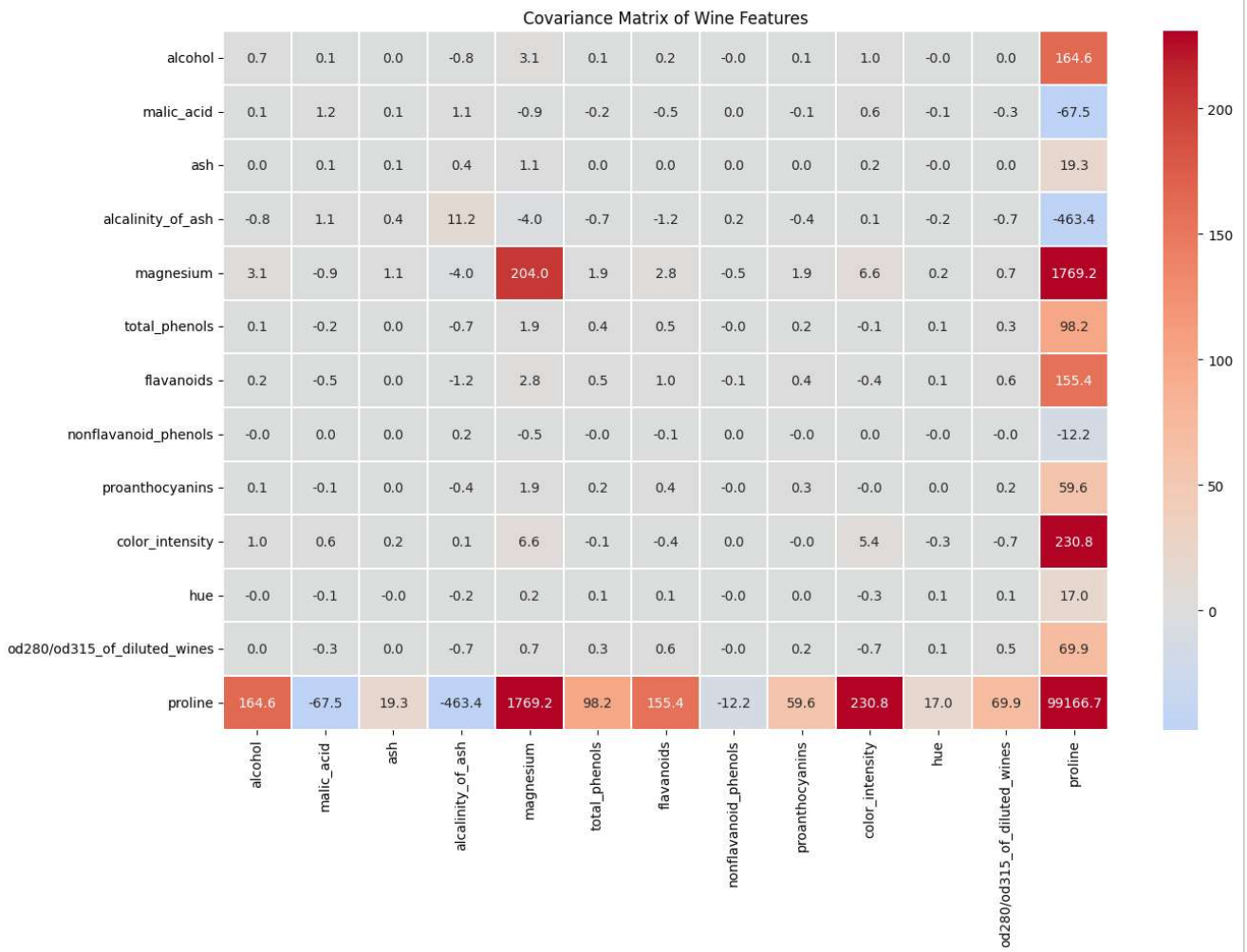
sns.heatmap(
    covariance_matrix,
    cmap="coolwarm",
    center=0,
    robust=True,
    annot=True,
    fmt=".1f",
    linewidths=0.3
)

plt.title("Covariance Matrix of Wine Features")
plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()

print("Covariance matrix calculated successfully!")
```

```
Covariance Matrix:
```

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_
alcohol	0.66	0.09	0.05	-0.84	3.14	0.15	0.19	
malic_acid	0.09	1.25	0.05	1.08	-0.87	-0.23	-0.46	
ash	0.05	0.05	0.08	0.41	1.12	0.02	0.03	
alcalinity_of_ash	-0.84	1.08	0.41	11.15	-3.97	-0.67	-1.17	
magnesium	3.14	-0.87	1.12	-3.97	203.99	1.92	2.79	
total_phenols	0.15	-0.23	0.02	-0.67	1.92	0.39	0.54	
flavanoids	0.19	-0.46	0.03	-1.17	2.79	0.54	1.00	
nonflavanoid_phenols	-0.02	0.04	0.01	0.15	-0.46	-0.04	-0.07	
proanthocyanins	0.06	-0.14	0.00	-0.38	1.93	0.22	0.37	
color_intensity	1.03	0.64	0.16	0.15	6.62	-0.08	-0.40	
hue	-0.01	-0.14	-0.00	-0.21	0.18	0.06	0.12	
od280/od315_of_diluted_wines	0.04	-0.29	0.00	-0.66	0.67	0.31	0.56	
proline	164.57	-67.55	19.32	-463.36	1769.16	98.17	155.45	




```
Covariance matrix calculated successfully!
```

Interpretation:

The covariance matrix calculates the relationship between all 13 numerical features of the Wine dataset. Positive covariance indicates that two features tend to move in the same direction, whereas negative covariance indicates that they tend to move in opposite directions. The heatmap provides a visual representation of these relationships. It helps identify how different numerical features vary together.

```
# STEP 5 - DATA SCALING

scaler = StandardScaler()

# Apply standardization
scaled_data = scaler.fit_transform(features)

# Convert scaled data into DataFrame
scaled_df = pd.DataFrame(
    scaled_data,
    columns=wine.feature_names
)

print("Data scaling completed successfully!")

print("\nFirst five rows after scaling:")
display(scaled_df.head().round(3))

print("\nMean of scaled features (first five):")
print(scaled_df.mean().head().round(4))

print("\nStandard deviation of scaled features (first five):")
print(scaled_df.std(ddof=0).head().round(4))
```

```
Data scaling completed successfully!

First five rows after scaling:
```

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	proanthocyanins
0	1.519	-0.562	0.232	-1.170	1.914	0.809	1.035	-0.660	1.225
1	0.246	-0.499	-0.828	-2.491	0.018	0.569	0.734	-0.821	-0.545
2	0.197	0.021	1.109	-0.269	0.088	0.809	1.216	-0.498	2.136
3	1.692	-0.347	0.488	-0.809	0.931	2.491	1.467	-0.982	1.032
4	0.296	0.228	1.840	0.452	1.282	0.809	0.663	0.227	0.401

```
Mean of scaled features (first five):
alcohol              0.0
malic_acid           0.0
ash                 -0.0
alcalinity_of_ash   -0.0
magnesium           -0.0
dtype: float64

Standard deviation of scaled features (first five):
alcohol              1.0
malic_acid           1.0
ash                  1.0
alcalinity_of_ash    1.0
magnesium            1.0
dtype: float64
```

Interpretation:

Data scaling was successfully performed using StandardScaler. The technique transforms the numerical features so that they have approximately zero mean and unit standard deviation. This process is important because the Wine dataset contains features with different numerical ranges. Standardization ensures that features with larger values do not dominate the PCA analysis.

```
# STEP 6 - PRINCIPAL COMPONENT ANALYSIS

# Apply PCA with two components
pca = PCA(n_components=2)

principal_components = pca.fit_transform(scaled_data)

# Create a DataFrame
pca_df = pd.DataFrame(
    principal_components,
    columns=[
        "Principal Component 1",
        "Principal Component 2"
    ]
)
```

```
# Add class labels
pca_df["Class_Name"] = data["Class_Name"].values

print("PCA completed successfully!")

# Explained variance
print("\nExplained variance ratio:")

print(
    "Principal Component 1:",
    f"{pca.explained_variance_ratio_[0]:.4f}"
)

print(
    "Principal Component 2:",
    f"{pca.explained_variance_ratio_[1]:.4f}"
)

total_variance = pca.explained_variance_ratio_.sum()

print(
    "Total variance explained by two components:",
    f"{total_variance:.4f}"
)

print(
    "Total variance explained (%):",
    f"{total_variance * 100:.2f}%"
)

# PCA visualization
plt.figure(figsize=(9, 6))

sns.scatterplot(
    data=pca_df,
    x="Principal Component 1",
    y="Principal Component 2",
    hue="Class_Name",
    palette="Set1",
    s=70
)

plt.title("PCA - Wine Class Separation")

plt.xlabel(
    f"PC1 ({pca.explained_variance_ratio_[0] * 100:.2f}% variance)"
)

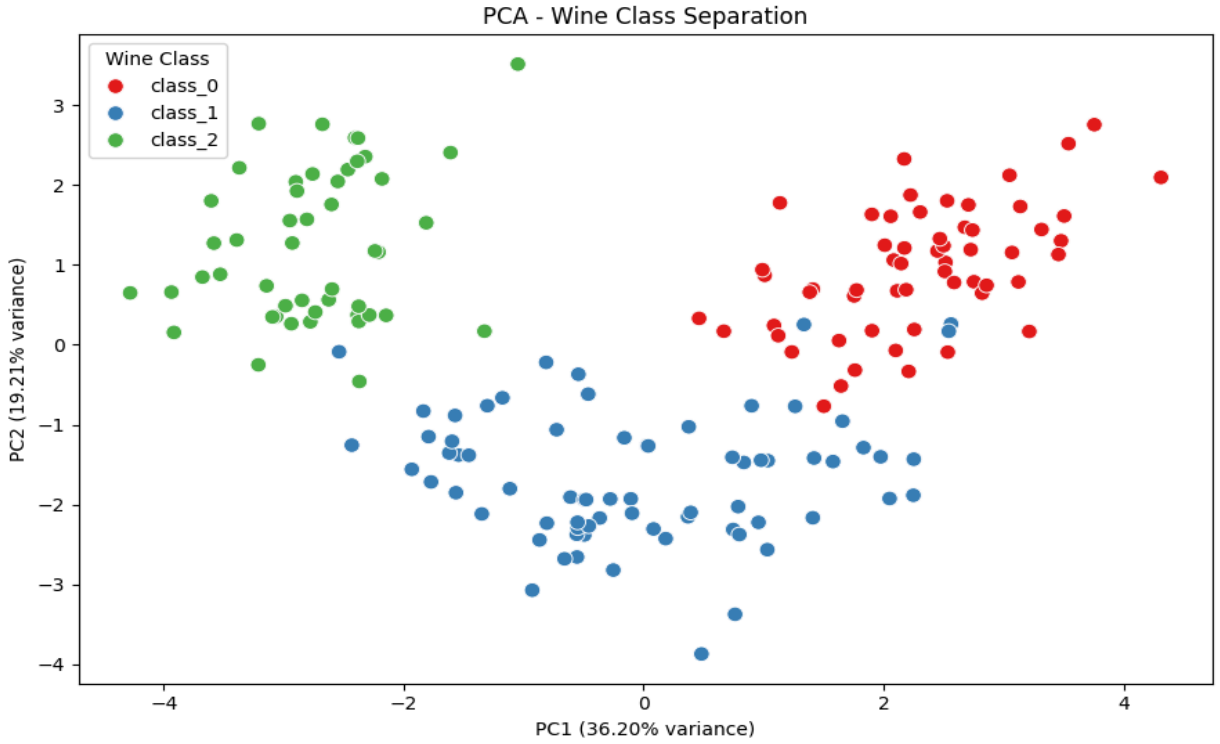
plt.ylabel(
    f"PC2 ({pca.explained_variance_ratio_[1] * 100:.2f}% variance)"
)

plt.legend(title="Wine Class")
plt.tight_layout()
plt.show()
```

```
PCA completed successfully!
```

```
PCA completed successfully!

Explained variance ratio:
Principal Component 1: 0.3620
Principal Component 2: 0.1921
Total variance explained by two components: 0.5541
Total variance explained (%): 55.41%
```



Interpretation- PCA was successfully applied to reduce the original 13 numerical features into two principal components. The first principal component explains 36.20% of the total variance, while the second principal component explains 19.21%. Together, these two components explain 55.41% of the total variance. The PCA scatterplot helps visualize the distribution and separation of the three wine classes in a two-dimensional space.

Conclusion- The Wine dataset was successfully analysed using Pandas, Matplotlib, Seaborn, and Scikit-learn. Basic statistics were calculated, boxplots and a scatterplot were generated, the covariance matrix was visualized, and the data was standardized. PCA reduced the 13 numerical features to two principal components. These components explained 55.41% of the total variance, allowing the three wine classes to be visualized in two dimensions.